IT Project Management

Lecture 2:
Project Process, Lean, and Agile

Dr. Hung-Nghiep Tran
nghiepth@uit.edu.vn

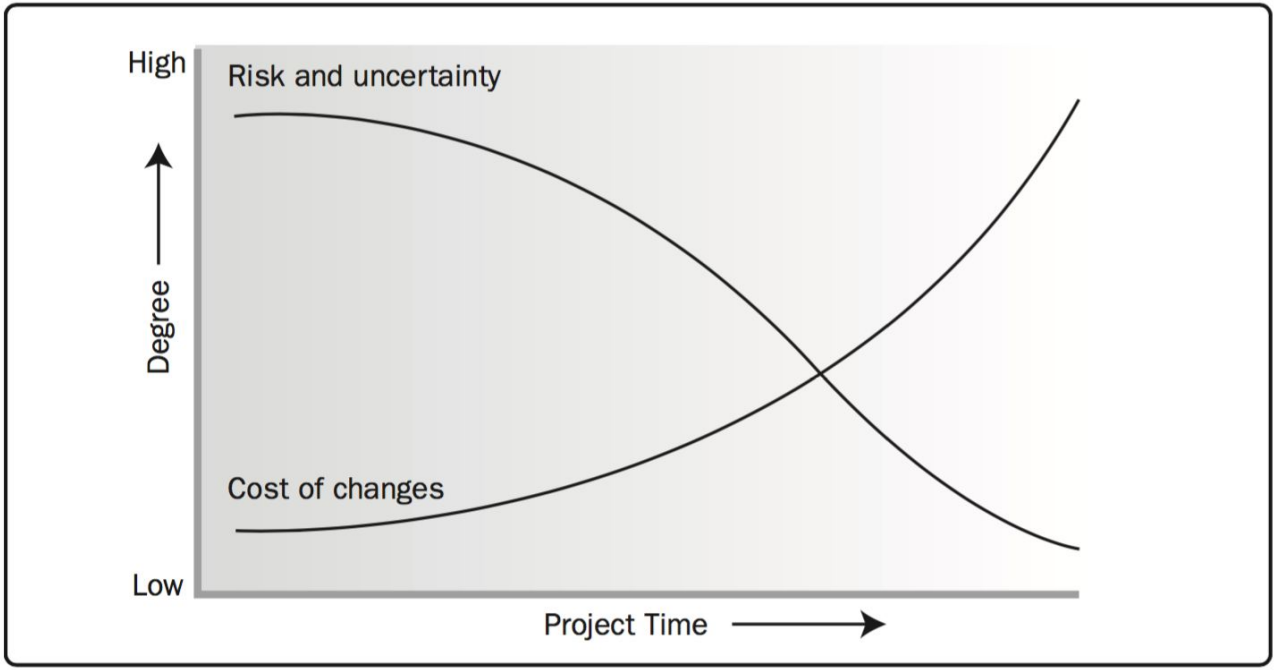
University of Information Technology, VNU-HCM, Vietnam
2025-2026

About this lecture

- Learning objectives:
  - Understand the basics of IT project management process
  - Differentiate between traditional and modern methodologies
  - Explain key principles of Lean and Agile approaches

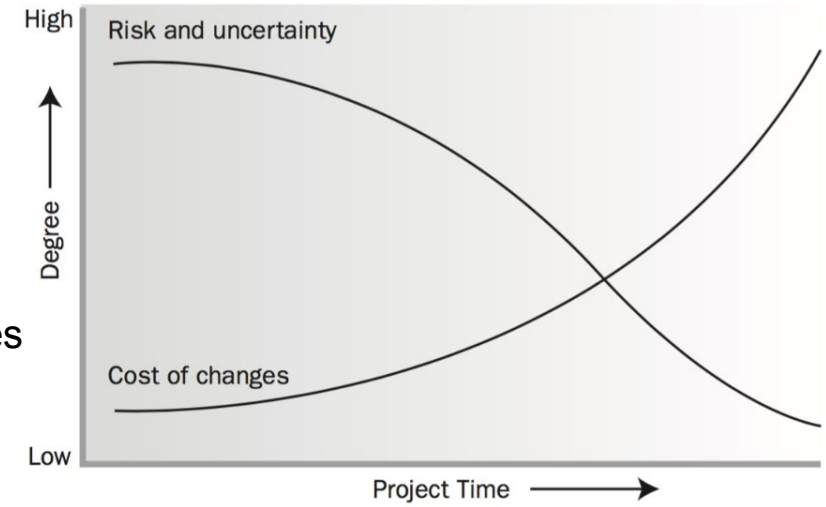
IT project management process

Project Life Cycle



Project Life Cycle

- In early phases
  - resource needs are usually lowest
  - the level of uncertainty (risk) is highest
  - more easily influence changes in the project
- In middle phases
  - the certainty of completing a project improves
  - more resources are needed
- The final phase
  - ensuring that project requirements were met
  - the sponsor approves completion of the project

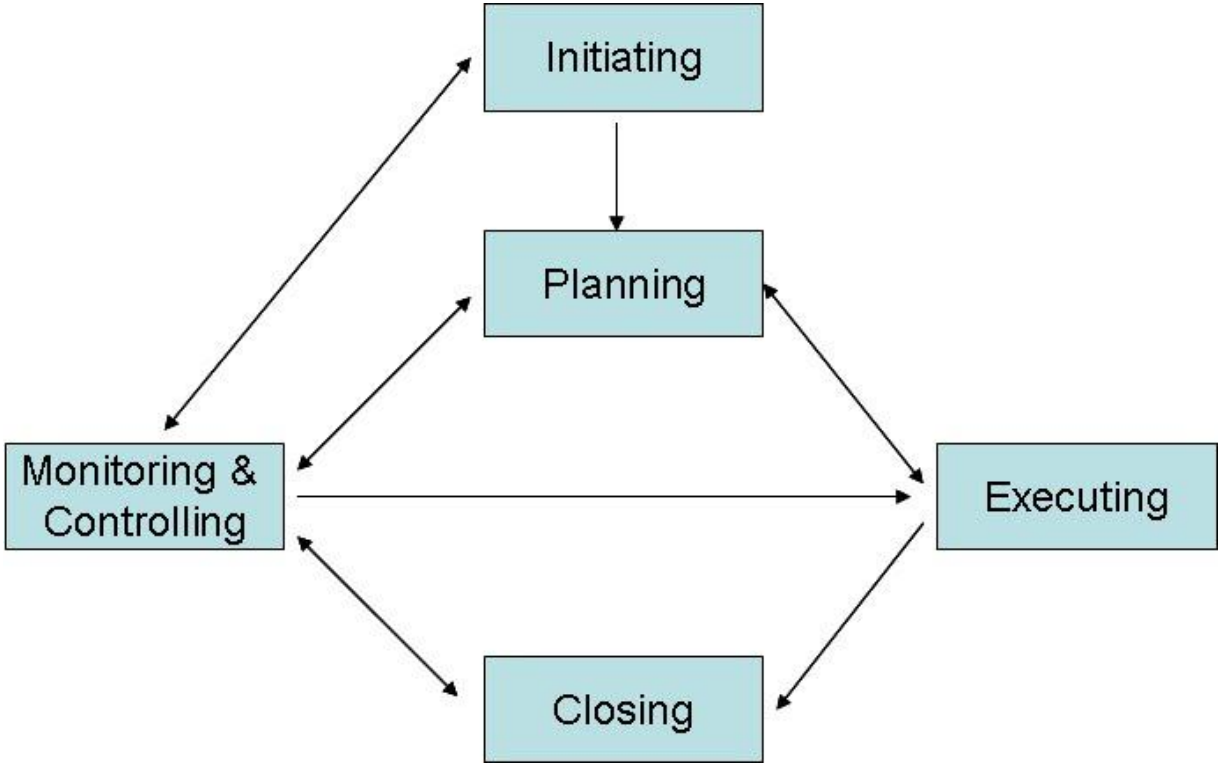


5 process groups of a Project (PMBOK Guide)

5 process groups (phases):

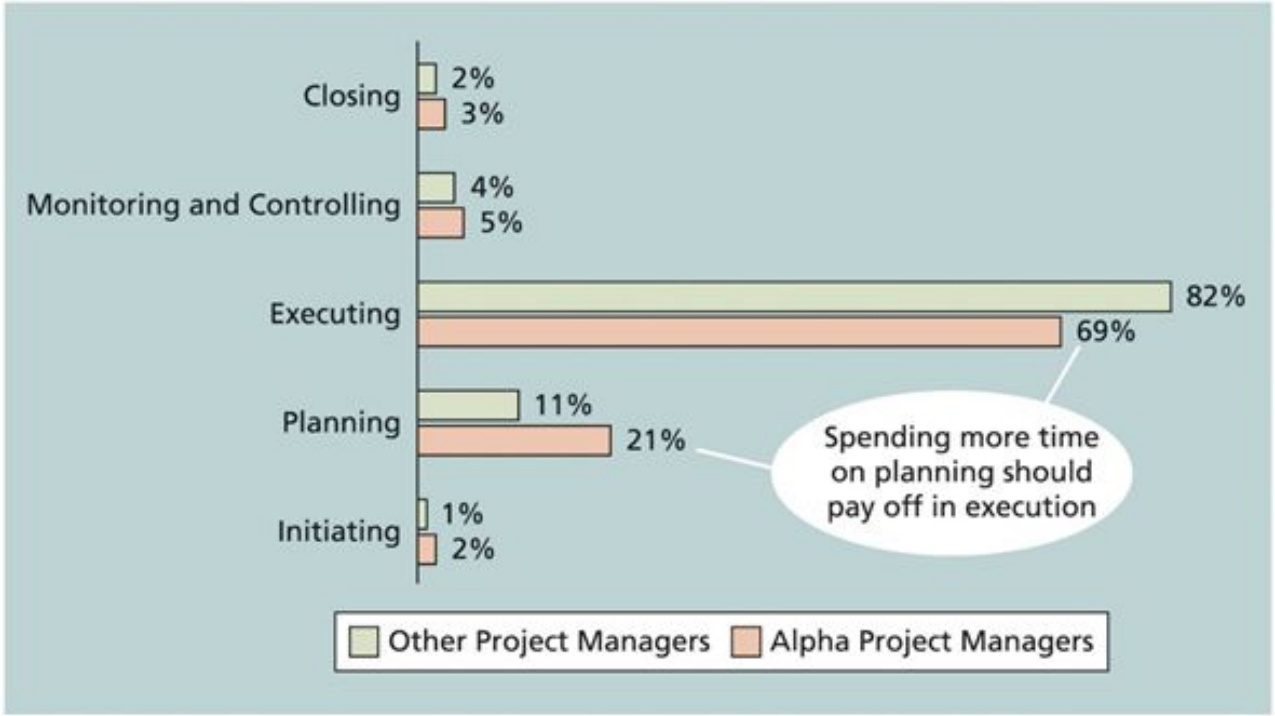
- **Initiating** – Define goals, stakeholders
- **Planning** – Scope, timeline, budget
- **Executing** – Development, testing
- **Monitoring & Controlling** – Track progress, adjust
- **Closing** – Deliver & evaluate

5 process groups of a Project (PMBOK Guide)



5 process groups of a Project (PMBOK Guide)

5 process groups of a Project (PMBOK Guide)



Source: Andy Crowe

Percentage of time spent on each process group

5 process groups of a Project (PMBOK Guide)

Process Group	Description	Goal	Outcome
Initiating	Starting the project	Authorize the project	Project charter
Planning	Organizing and preparing	Plan and schedule the work to perform the project	Project management plan
Executing	Carrying out the work	Perform the project work	Project deliverables: product, service, results
Monitoring and Controlling	Spans the project lifecycle	Monitor the progress of the project to identify variances and correct them	Change requests and recommendations for preventive and corrective actions
Closing	Closing the project formally	Close the project	Product acceptance, contract closure, and archiving

5 process groups of a Project (PMBOK Guide)

QUIZ

Match each task to the correct process group:

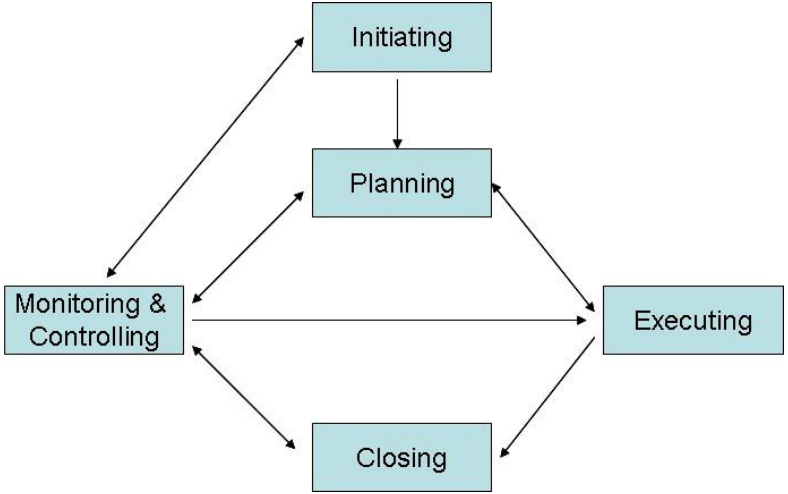
- Measure delivery checkpoints
- Writing project requirements
- Unit testing software
- Holding a stakeholder meeting
- Reviewing project success

5 process groups of a Project (PMBOK Guide)

QUIZ

Match each task to the correct process group:

- Measure delivery checkpoints
- Writing project requirements
- Unit testing software
- Holding a stakeholder meeting
- Reviewing project success

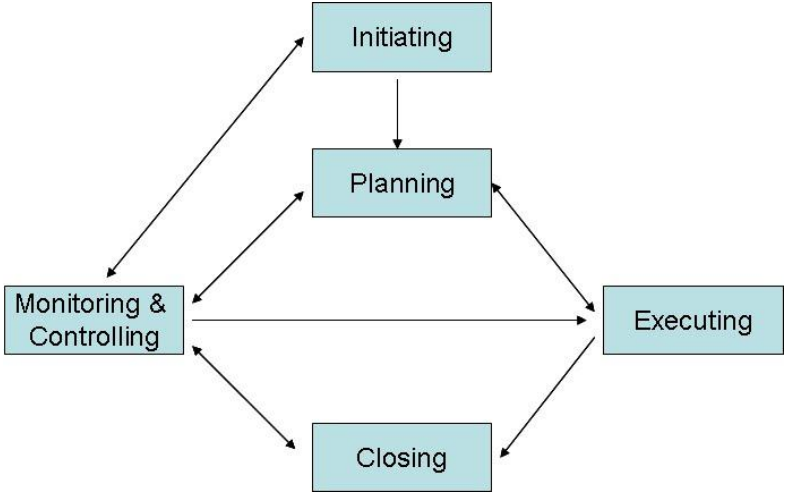


5 process groups of a Project (PMBOK Guide)

QUIZ

Match each task to the correct process group:

- Measure delivery checkpoints
  → Monitoring & Controlling
- Writing project requirements
  → Planning
- Unit testing software
  → Executing
- Holding a stakeholder meeting
  → Initiating
- Reviewing project success
  → Closing



Technical details of Software Development Life Cycle: 7 stages of an IT project

1. Initiation/Identification
2. Analysis
3. Design
4. Implementation
5. System Testing
6. User Acceptance Testing - UAT
7. Operation & Maintenance

Technical details: 7 stages of an IT project

Initiation/Identification

- Define the project goals, objectives, and scope
- Identify key stakeholders and their needs
- Conduct feasibility studies and business case analysis

- Example: A company decides to build a mobile banking app

Technical details: 7 stages of an IT project

Analysis

- Gather user and business requirements
- Conduct risk assessment and technical feasibility study
- Identify constraints (scope, cost, time)

- Example: Surveying bank customers to determine needed app features

Technical details: 7 stages of an IT project

Design

- Create system architecture and database design
- Define user interfaces (UI/UX)
- Plan infrastructure & security measures

- Example: Designing the app layout, security encryption, and backend system

Technical details: 7 stages of an IT project

Implementation

- Developers write the code and integrate system components
- Implement backend, frontend, and database systems
- Follow Agile/Scrum/Kanban/Lean, or Waterfall, or DevOps methodologies

- Example: Engineers coding the banking app and setting up cloud servers

Technical details: 7 stages of an IT project

System Testing

- Conduct system tests, integration tests, and performance tests
- Fix bugs and security vulnerabilities
- Ensure the system works across different devices and environments

- Example: Testing if the banking app processes transactions correctly

Technical details: 7 stages of an IT project

User Acceptance Testing - UAT

- End-users test the system in real-world conditions
- Verify if the project meets business and user requirements
- Get final approval before launch

- Example: A group of bank employees test the app before public release

Technical details: 7 stages of an IT project

Operation & Maintenance

- Deploy the system to production (go live)
- Provide user training and documentation
- Monitor for bugs, security threats, and performance issues
- Regularly update and optimize the system

- Example: The bank officially launches the app and monitors transactions

Connections

7 stages of an IT project vs. 5 PMI process groups

7 IT Project Stages	PMI's 5 Process Groups	Description
Initiation/Identification	Initiating	Define project goals, stakeholders, and feasibility
Analysis	Planning	Gather requirements, define scope, identify risks
Design	Planning	Create system architecture, UI/UX, security
Implementation	Executing	Develop and build the IT system
System Testing	Monitoring & Controlling	Ensure quality, test performance, fix bugs
User Acceptance Testing	Monitoring & Controlling	Verify with end-users, get approval
Operation & Maintenance	Closing	Deploy the system, provide support & updates

7 stages of an IT Project

QUIZ

Match each task to the correct project stage:

- Measure delivery checkpoints
- Writing project requirements
- Unit testing software
- Holding a stakeholder meeting
- Reviewing project success

7 stages of an IT Project

QUIZ

Match each task to the correct project stage:

- Measure delivery checkpoints
- Writing project requirements
- Unit testing software
- Holding a stakeholder meeting
- Reviewing project success

7 IT Project Stages	**PMI's 5 Process Groups**
Initiation/Identification	Initiating
Analysis	Planning
Design	Planning
Implementation	Executing
System Testing	Monitoring & Controlling
User Acceptance Testing (UAT)	Monitoring & Controlling
Operation & Maintenance	Closing

7 stages of an IT Project

QUIZ

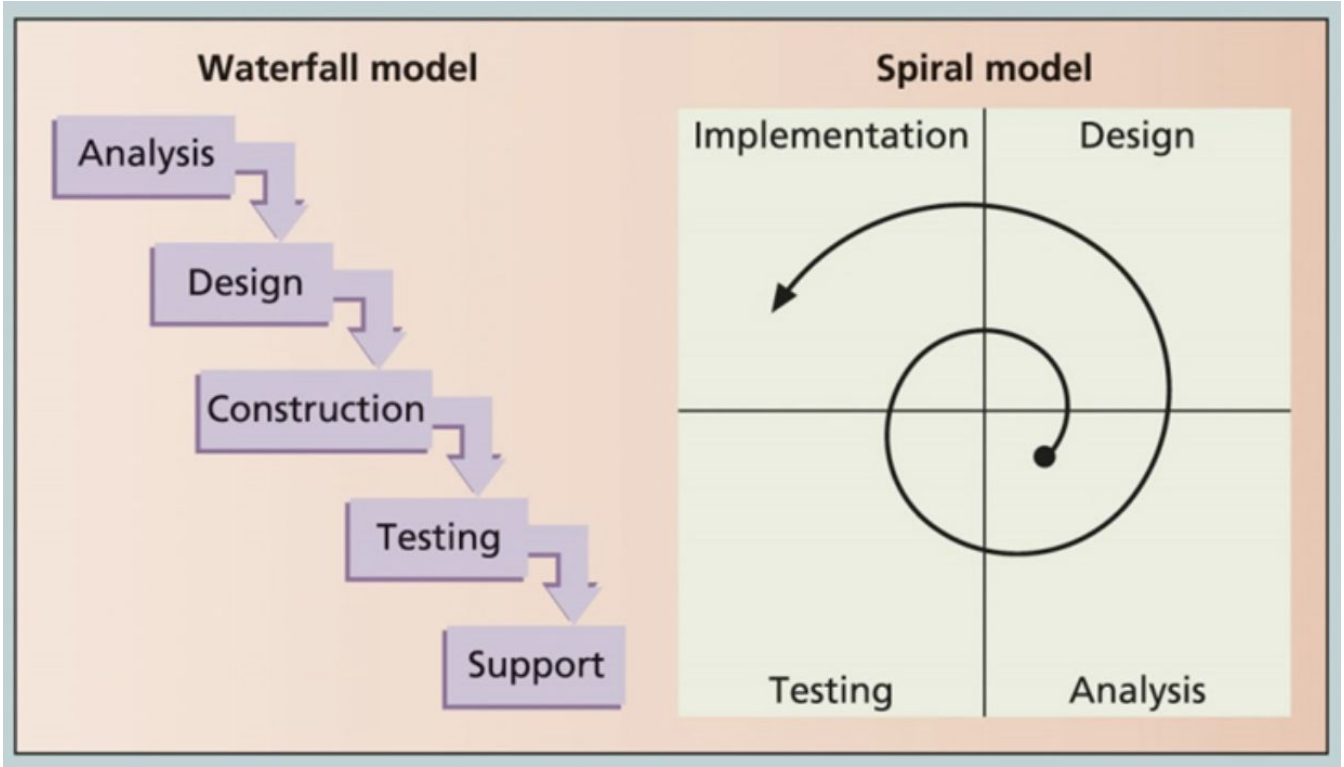
Match each task to the correct project stage:

- Measure delivery checkpoints
  → System Testing/UAT
- Writing project requirements
  → Analysis
- Unit testing software
  → Implementation
- Holding a stakeholder meeting
  → Initiation/Identification
- Reviewing project success
  → Operation & Maintenance

7 IT Project Stages	**PMI's 5 Process Groups**
Initiation/Identification	Initiating
Analysis	Planning
Design	Planning
Implementation	Executing
System Testing	Monitoring & Controlling
User Acceptance Testing (UAT)	Monitoring & Controlling
Operation & Maintenance	Closing

Project management approach

Traditional approach:
Predictive Life Cycle Models



Traditional approach:
Predictive Life Cycle Models

Traditional models:

- **Waterfall model:** well-defined, **linear stages** of systems development and support
- **Spiral model:** self-improving by **iterating through multiple stages** of development
- Prototyping model
- Rapid Application Development model
- Etc.

→ There is input from the beginning and only feedback at the end

no feedback from customers/users in the process

Traditional approach:
Predictive Life Cycle Models

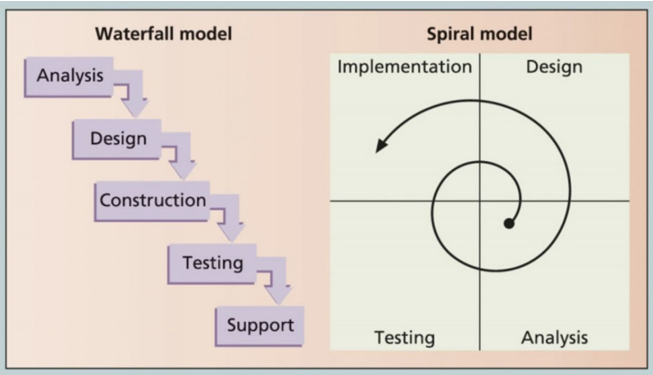
Predictive approach with **no feedback in the process**:

→ Suitable for traditional projects with **known, stable, fixed requirements**

- E.g., an airport construction project, a banking website

→ Not suitable for modern projects with more **complex**, **unknown,** or **continuously changing requirements**

- E.g., an AI chat bot application



Agile approach

What is Agile?

"Agile" means: to **move quickly and easily**

Agile is a flexible project management approach:

→ emphasizes **iteration, collaboration, and adaptability**

→ **requirements and solutions** evolve **collaboratively**

Agile

The Agile Manifesto (2001)

4 Key Values:

- **Individuals & interactions** over processes & tools
- **Working software** over comprehensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a plan

Agile

The Agile Manifesto (2001)

12 Principles:

1. Satisfy customers through **early and continuous delivery**
2. **Welcome changing requirements**, even late in development
3. Deliver **working software frequently** (weeks instead of months)
4. **Business & development teams must work together daily**
5. Build projects around **motivated individuals**
6. Use **face-to-face communication** whenever possible

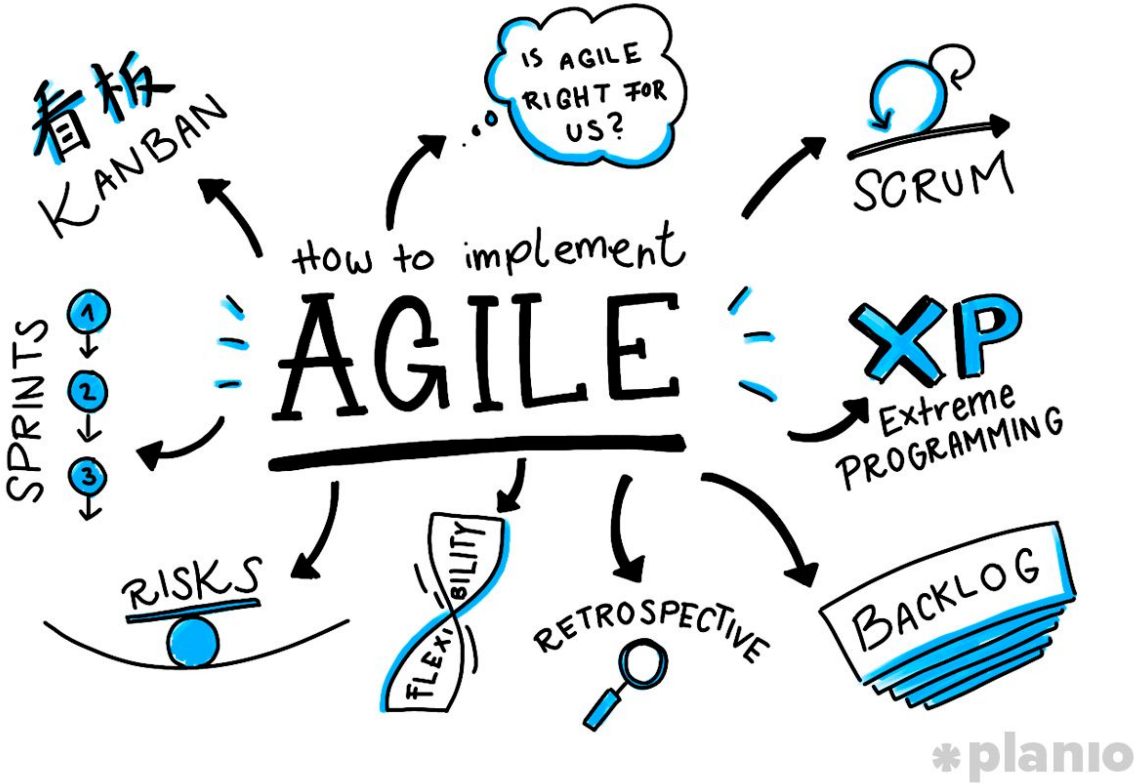
Agile

The Agile Manifesto (2001)

12 Principles:

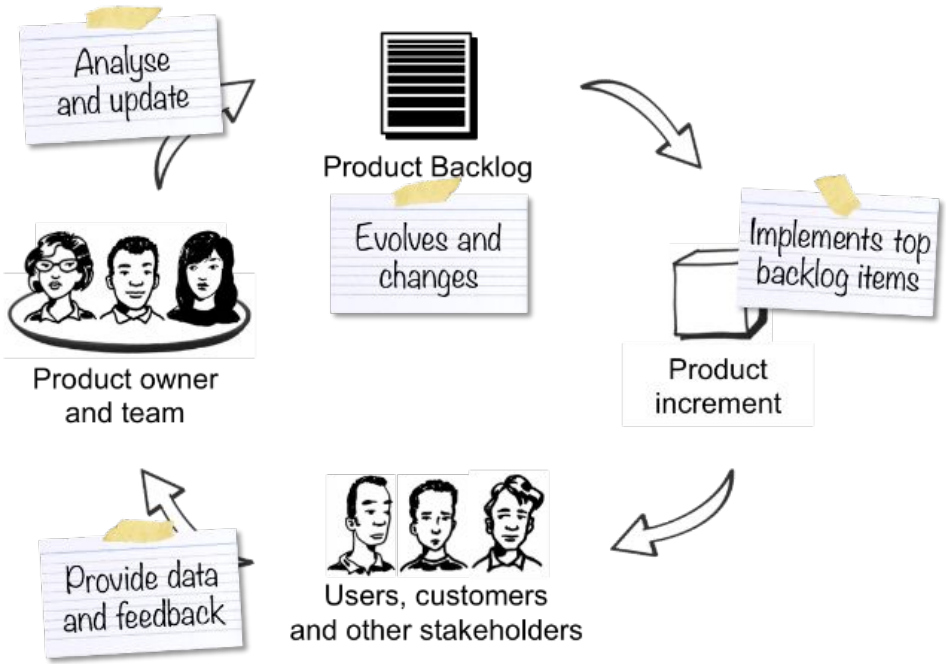
7. **Working software is the primary measure of success**
8. Maintain a **sustainable work pace** (no overloading teams)
9. Focus on **technical excellence & good design**
10. Maximize **simplicity** – do only necessary work
11. **Self-organizing teams** deliver the best results
12. Regularly reflect on how to **become more effective**

Agile


看板
KANBAN
IS AGILE RIGHT FOR US?
SCRUM
how to implement
AGILE
SPRINTS
1
2
3
XP
Extreme PROGRAMMING
RISKS
FLEXIBILITY
RETROSPECTIVE
BACKLOG
planio

Scrum

- Scrum is the **most widely used Agile framework** for complex IT projects
- **Scrum focuses on the process**
  - Product owner, scrum master, scrum team
  - Sprint, meeting, review, retrospective
- The key features of Scrum are:
  - **Sprint**: Time-boxed cycles, usually 2-4 weeks, to finish a feature
  - **Story**: (user/goal-based) motivation for each feature



Scrum

A Scrum team consists of **3 main roles**:

- **Product Owner**

  → **“Captain”:** Defines & prioritizes the work

- **Scrum Master**

  → **“Cheerleader”/“MC”:** Facilitates the process, removes blockers

- **Scrum Team (Development Team)**

  → **“Main force”:** Cross-functional team of 5 to 9 people (BA, Dev, Tester)

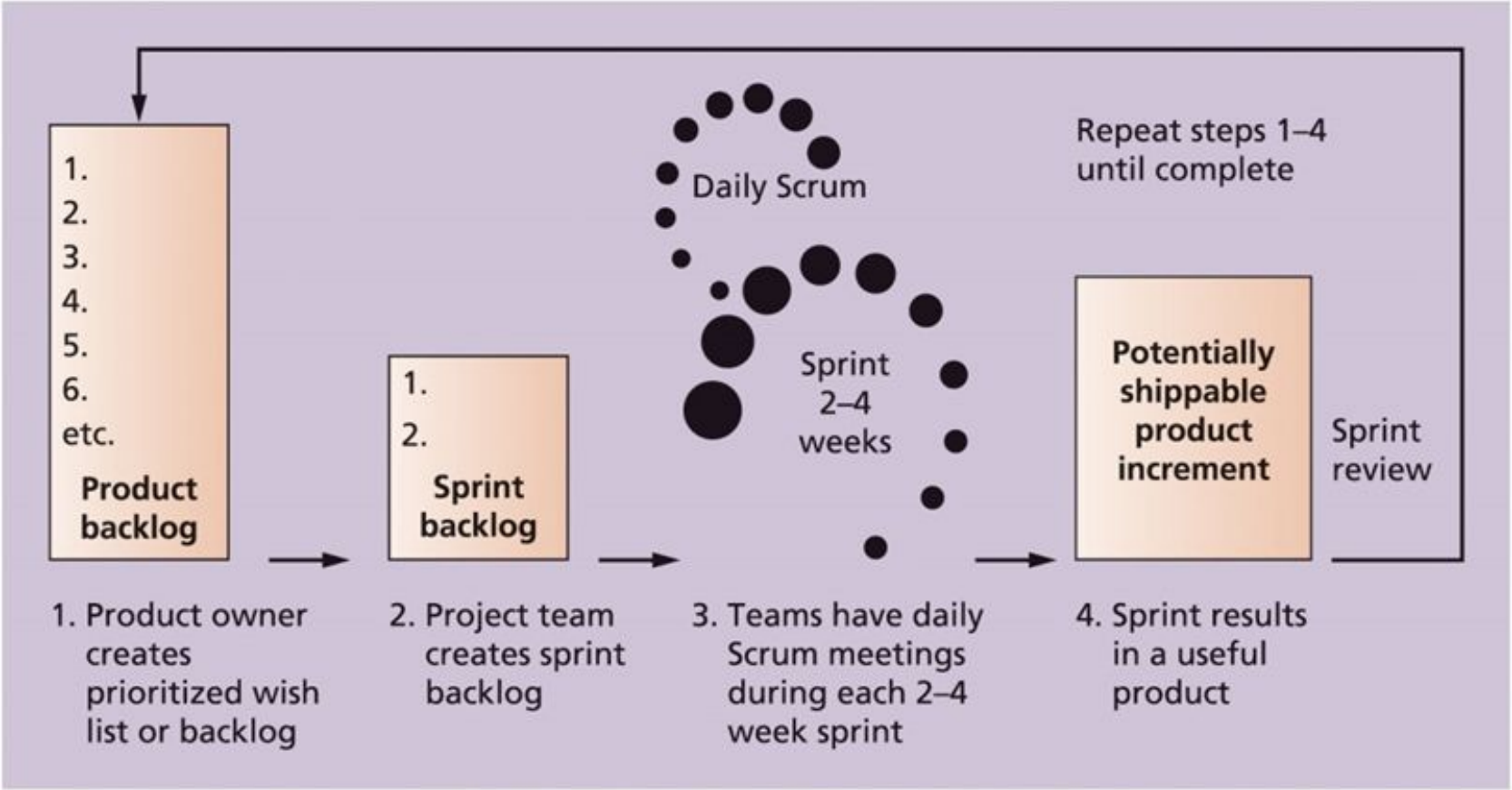
  → Builds & delivers the product in each sprint

Scrum

Scrum workflow includes:

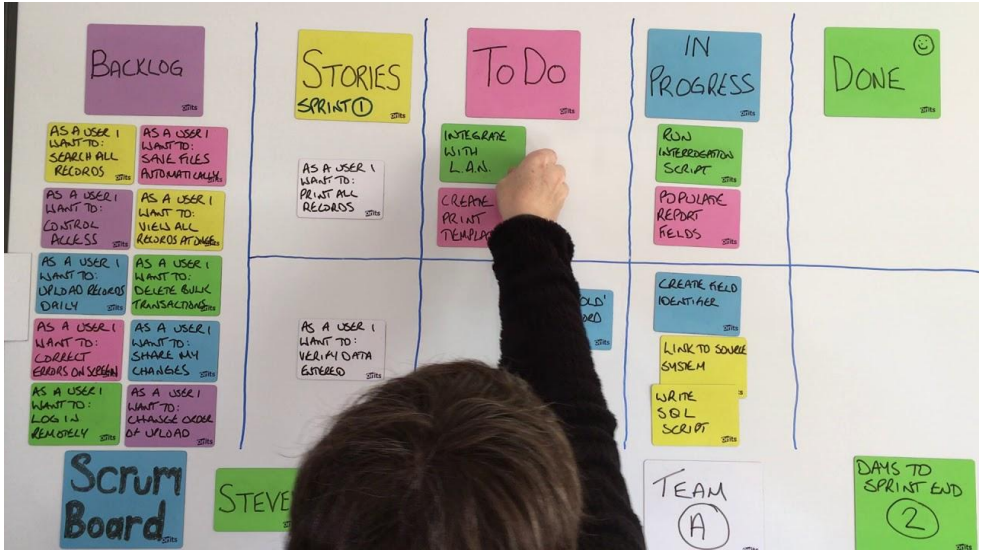
- **Sprint Planning** → Team plans work for the next sprint
- **Daily Stand-up (15 min)** → Short meeting daily to discuss progress & blockers
- **Sprint Review** → Demo completed work to stakeholders at the end of sprint
- **Sprint Retrospective** → Reflect & improve for the next sprint

Scrum


1.
2.
3.
4.
5.
6.
etc.
Product backlog
1.
2.
Sprint backlog
Daily Scrum
Sprint 2–4 weeks
Repeat steps 1–4 until complete
Potentially shippable product increment
Sprint review
1. Product owner creates prioritized wish list or backlog
2. Project team creates sprint backlog
3. Teams have daily Scrum meetings during each 2–4 week sprint
4. Sprint results in a useful product

Kanban

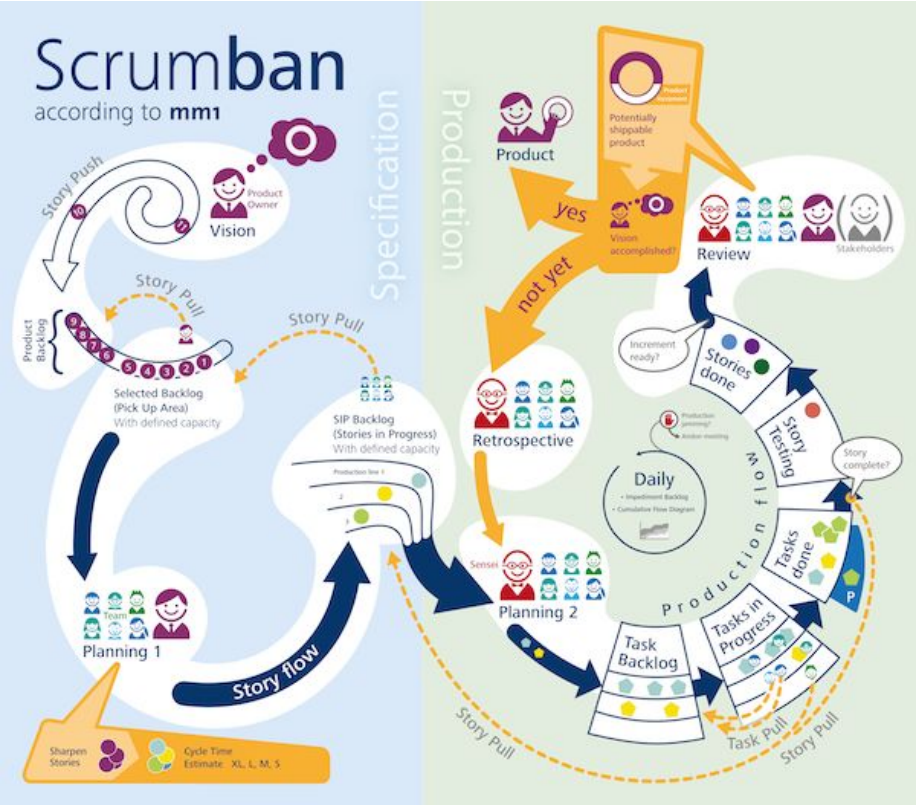
- Developed by Toyota Motor Corporation
- **Kanban focuses on visualization**
  - Uses visual cues to guide workflow optimization
  - Kanban cards show new work, work in progress, and work completed
- The key features of Kanban are:
  - **Workflow tracking**: manage flow in real-time, focus on finishing tasks without overload
  - **Story**: (user/goal-based) motivation for each feature



Scrum & Kanban

Scrum and Kanban can be **used together**:

- Scrum focuses on the **process**
  - Product owner, scrum master, scrum team
  - Sprint, meeting, review, retrospective
- Kanban focuses on **visualization**
  - Uses visual cues to guide workflow optimization
  - Kanban cards show new work, work in progress, and work completed



Extreme Programming (XP) principles

- XP focuses on **engineering best practices** to ensure **high-quality software**:
  - **Test-Driven Development** (TDD) → Write tests before coding
  - **Pair Programming** → Two developers write code together
  - **Continuous Integration** (CI/CD) → Deploy small updates frequently
  - **Refactoring** → Improve code without changing functionality

- XP is useful for:
  - Mission-critical software (test, etc.)
  - Frequent releases (CI/CD, etc.)

Case Study: How Agile Helped a Software Startup Scale

- Company: **Spotify**
- Problem:
  - As Spotify grew to ~1 million users:
    - Its traditional project management approach **couldn't scale**
    - Teams were slow to adapt to new features & market trends

Case Study: How Agile Helped a Software Startup Scale

- **Agile Approach** (Spotify Model)**:**
  - "Squads" → Autonomous teams working like **mini-startups**
    (Cross-functional: BA, Dev, Tester. Small: 5-9 people)
  - **Tribes & Chapters** → Teams collaborate but remain flexible
  - **Short Iterations & Continuous Feedback**

- Results:
  - Spotify scaled from **1M to 400M+ users**
  - New features rolled out **10x faster**

Agile, Scrum & Kanban

QUIZ

Imagine you are developing an AI chatbot for a specific domain.

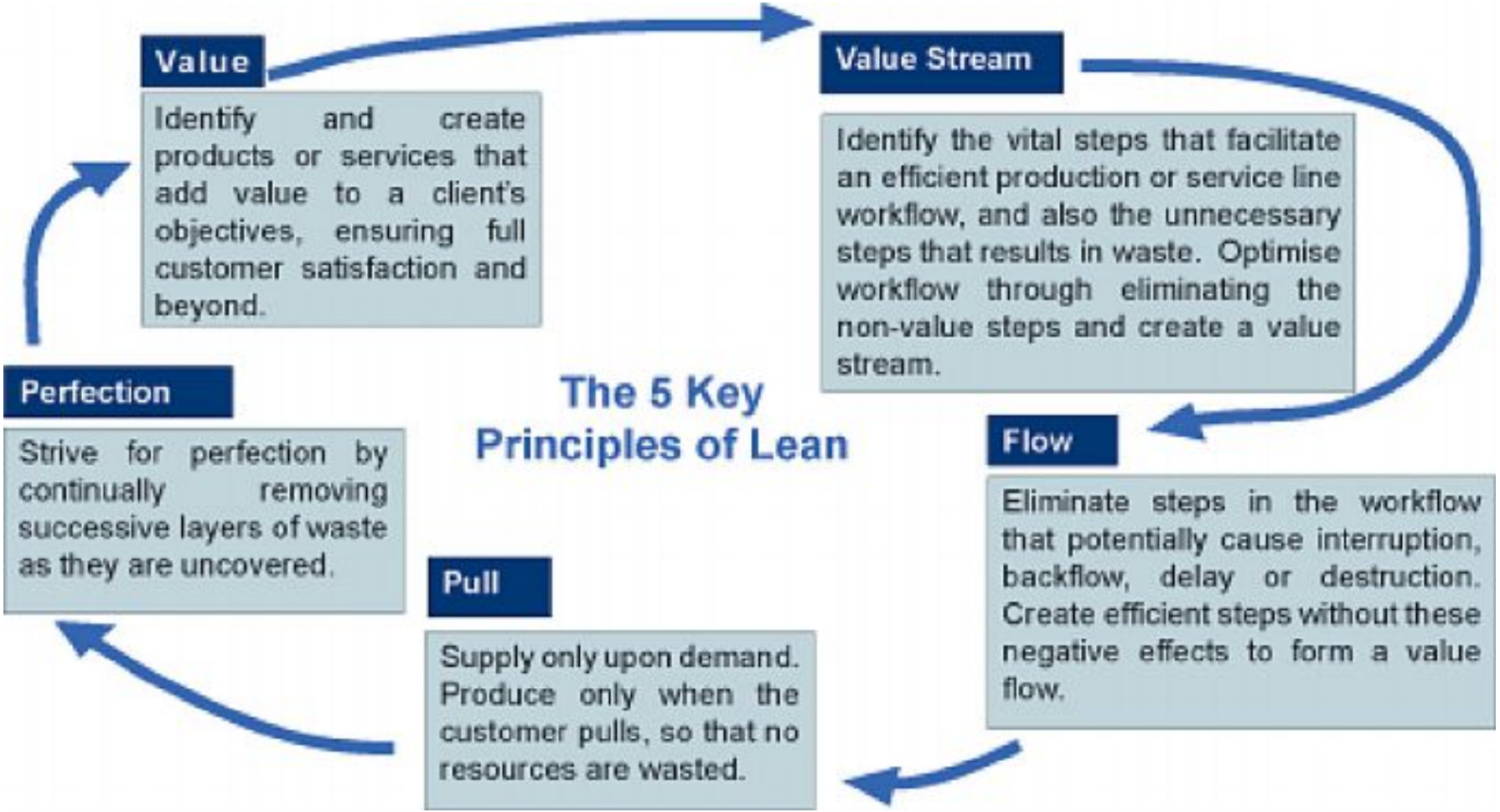
- How would you do it with Agile?
  - Role playing product owner, scrum master, scrum/dev team
  - Visualize task progress, tell stories

Lean approach

What is Lean?

- Originated from Toyota Production System for manufacturing
- Focus:
  - Efficiency
  - Eliminating waste
  - Delivering maximum value with minimum resources

What is Lean?


Value
Identify and create products or services that add value to a client's objectives, ensuring full customer satisfaction and beyond.
Value Stream
Identify the vital steps that facilitate an efficient production or service line workflow, and also the unnecessary steps that results in waste. Optimise workflow through eliminating the non-value steps and create a value stream.
The 5 Key Principles of Lean
Flow
Eliminate steps in the workflow that potentially cause interruption, backflow, delay or destruction. Create efficient steps without these negative effects to form a value flow.
Pull
Supply only upon demand. Produce only when the customer pulls, so that no resources are wasted.
Perfection
Strive for perfection by continually removing successive layers of waste as they are uncovered.

Lean in IT Projects: 7 principles

1. **Eliminate Waste**
2. Built-in Quality
3. Create Knowledge
4. Delay Commitment
5. Deliver Fast
6. Empower People
7. Continuous Improvement

Lean in IT Projects: 7 principles

Eliminate Waste

- Waste (not adding value) should be removed from the project
  - Examples of Waste in IT Projects:
    - Unnecessary features
    - Waiting time (for approvals, feedback, etc.)
    - Rework (fixing mistakes, unclear requirements)
    - Task switching (context switching cost)

Solution:

- Use **Minimum Viable Product (MVP)** to launch essential features first
  - MVP is the most simple product that can still demonstrate the key features
  - Follow Pareto law: 20% efforts for 80% results

Lean in IT Projects: 7 principles

Built-in Quality

- Quality should be **integrated into the process**, not added at the end
- Bugs and mistakes should be **prevented**, not just fixed after testing

Solutions:

- Automated Testing → Run frequent tests early in development
- Code Reviews → Ensure quality before merging changes
- Continuous Integration (CI/CD) → Deploy software automatically & frequently

Lean in IT Projects: 7 principles

Create Knowledge (Continuous Learning)

- Teams should **continuously learn** and improve by documenting lessons learned and best practices
- Encourage knowledge sharing through **wikis, retrospectives, and documentation**

Solutions:

- Pair Programming → Developers work together, share skills
- Retrospectives → Regular meetings to reflect and improve
- Technical Documentation → Prevents knowledge loss

Lean in IT Projects: 7 principles

Delay Commitment (Decide as Late as Possible)

- Don't make **big decisions too early** when uncertainty is high
- Keep options open and **gather more information** before making commitments

Example:

- A startup shouldn’t commit to a **single technology stack** before testing different options on small prototypes

Lean in IT Projects: 7 principles

Deliver Fast (Optimize Flow & Reduce Cycle Time)

- Work should move through the process **smoothly and quickly**
- Reduce cycle **time from idea to deployment**

Example:

- Agile sprints (2-4 weeks) instead of long 6-month waterfall plans
- Kanban boards to track progress and prevent bottlenecks

Lean in IT Projects: 7 principles

Empower People

- Enable teams to make decisions and **improve their own processes**
- **Reduce micromanagement** and encourage collaboration

Solutions:

- Cross-functional teams → Developers, testers, and business analysts work together
- Self-organizing teams → Agile teams choose how they work

Lean in IT Projects: 7 principles

Continuous Improvement (Kaizen)

- Small, **continuous improvements over time** create big efficiency gains
- Regular **feedback loops** help refine processes and remove inefficiencies

Examples:

- Sprint Retrospectives → Teams reflect and improve after each sprint
- Daily Stand-ups → Quick check-ins to identify blockers and improvements

Case Study: How Lean Helped a Software Startup

- Company: **Dropbox**

- Problem:
  - Initially built too many extra features before knowing what customers really wanted
  - Delays due to long development cycles
  - High operational costs

Case Study: How Lean Helped a Software Startup

- **Lean Approach:**
  - Developed an **MVP** → Released a basic version of Dropbox with only essential features first
  - Customer Feedback → Data driven analysis to decide on features
  - Automated testing & deployment → Faster releases, faster feedback

- **Results:**
  - Dropbox grew from 100,000 to 4 million users in 15 months
  - Focused only on high-value features
  - Reduced costs & eliminated wasted effort

Lean in IT Projects

QUIZ

Identify types of waste and suggest how Lean can improve

Example Scenarios:

1. A team building an e-commerce website but developing features nobody asked for
2. A government IT project that spends months writing detailed documentation before coding begins
3. A company where developers keep getting interrupted by last-minute feature requests

Lean in IT Projects

QUIZ

1. A team building an e-commerce website but developing features nobody asked for

→ Unnecessary Features ⇒ Build and launch an MVP first

2. A government IT project that spends months writing detailed documentation before coding begins

→ Delays Due to Long Planning ⇒ Use Agile iterations instead

3. A company where developers keep getting interrupted by last-minute feature requests

→ Frequent Task Switching ⇒ Use Kanban to focus on fewer tasks at a time

Lean in IT Projects

QUIZ

Imagine you are developing an AI chatbot for a specific domain.

- How would you do it without lean?
- How would you do it with lean?

Lean in IT Projects

QUIZ

How would you apply Lean principles to your own university projects?

Lean vs. Agile vs. Waterfall

Lean vs. Agile vs. Waterfall

Feature	Lean	Agile	Waterfall
Focus	Efficiency	Flexibility	Predictability
Approach	Continuous	Iterative (Sprint)	Sequential
Feedback	Continuous	Frequent	At the end
Delivery	Quick improvements	Frequent small features	One final product
Project Length	Short (Early phase)	Iterative (Scaling phase)	Long
Flexibility	Medium	Adaptive	Rigid
Planning	Continuous	Ongoing	Heavy upfront
Documentation	Just enough	Minimal	Extensive
Best for	Efficiency-driven projects	Fast-changing projects	Large, stable projects

Summary

- You have learned:
  - Understand the basics of IT project management process
    → 5 process groups, 7 stages
  - Differentiate between traditional and modern methodologies
    → Feedback at the end vs. feedback in the process
  - Explain key principles of Lean and Agile approaches
    → Lean: eliminating waste. Agile: iterative improvements.
    → etc.

Thank you for listening

Q & A

Appendix

Case study

Case Study: Predictive Approach

JWD Consulting's Project Management Intranet Site

- This case study provides an example of what's involved in **initiating**, **planning**, **executing**, **controlling**, and **closing** an IT project

Case Study: Predictive Approach

Pre-Initiation and Initiation

- Initiating includes recognizing and starting a new project
  - Right kinds of projects for the right reasons
- Strategic planning should serve as the foundation for deciding which projects to pursue
  - Expresses the vision, mission, goals, objectives, and strategies of the organization
  - Provides the basis for IT project planning

Case Study: Predictive Approach

Pre-Initiation tasks

- It is good practice to lay the groundwork for a project before it starts
- Senior managers often perform several pre-initiation tasks
  - Determine the scope, time, and cost constraints for the project
  - Identify the project sponsor
  - Select the project manager
  - Develop a business case for a project
  - Meet with the PM to review the process and expectations for managing the project
  - Determine if the project should be divided into two or more smaller projects

Case Study: Predictive Approach

Initiation

Knowledge Area	Initiating Process	Initiating Process
Project Integration Management	Develop project charter	Project charter Assumption log
Project Stakeholder Management	Identify stakeholders	Stakeholder register Change requests Project management plan updates Project documents updates

Case Study: Predictive Approach

Initiation - Stakeholder register

Name	Position	Internal/External	Project Role	Contact Information
Joe Fleming	CEO	Internal	Sponsor	joe_fleming@jwdconsulting.com
Erica Bell	PMO Director	Internal	Project Manager	erica_bell@jwdconsulting.com
Michael Chen	Senior Consultant	Internal	Team Member	michael_chen@jwdconsulting.com
Kim Phuong	Business Analyst	External	Advisor	kim_phuong@client1.com
Louise Mills	PR Director	Internal	Advisor	louise_mills@jwdconsulting.com

Case Study: Predictive Approach

Initiation - Stakeholder Management Strategy

Name	Level of Interest	Level of Influence	Potential Management Strategies
Joe Fleming	High	High	Joe likes to stay on top of key projects and make money. Have a lot of short, face-to-face meetings and focus on achieving the financial benefits of the project.
Louise Mills	Low	High	Louise has a lot of things on her plate, and she does not seem excited about this project. She may be looking at other job opportunities. Show her how this project will help the company and her resume.

Case Study: Predictive Approach

Initiation

- Drafting the project charter
- Holding a project kick-off meeting
  - It's good practice to hold a kick-off meeting at the beginning of a project so that stakeholders can meet, review the goals of the project, and discuss future plans

Case Study: Predictive Approach

Initiation

Kick-Off Meeting
[Date of Meeting]

Project Name: Project Management Intranet Site Project

Meeting Objective: Get the project off to an effective start by introducing key stakeholders, reviewing project goals, and discussing future plans

Agenda:

- Introductions of attendees
- Review of the project background
- Review of project-related documents (business case and project charter)
- Discussion of project organizational structure
- Discussion of project scope, time, and cost goals
- Discussion of other important topics
- List of action items from meeting

Action Item	Assigned To	Due Date

Date and time of next meeting:

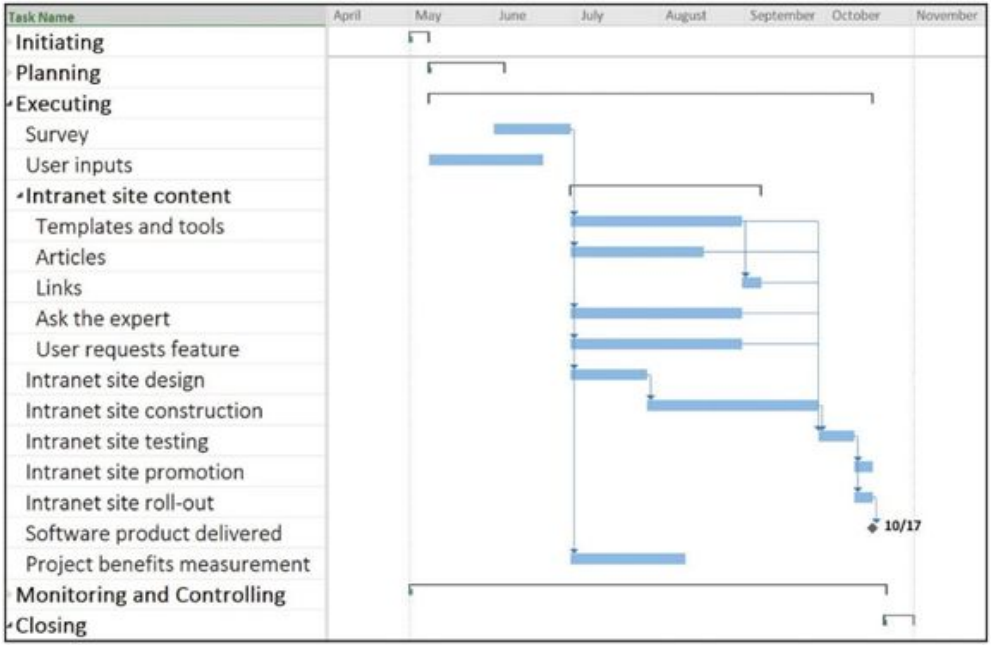
Case Study: Predictive Approach

Project Planning

- The main purpose of project planning is to guide execution
  - Every knowledge area includes planning information
- Key outputs included in the JWD project
  - Team contract
  - Project scope statement
  - Work breakdown structure (WBS)
  - Project schedule, in the form of a Gantt chart with all dependencies and resources entered
  - List of prioritized risks (part of a risk register)

Case Study: Predictive Approach

Project Planning



JWD Consulting intranet site project baseline Gantt chart

Case Study: Predictive Approach

Project Planning - list of prioritized Risks

Ranking	Potential Risk
1	Lack of inputs from internal consultants
2	Lack of inputs from client representatives
3	Security of new system
4	Outsourcing/purchasing for the article retrieval and Ask the Expert features
5	Outsourcing/purchasing for processing online payment transactions
6	Organizing the templates and examples in a useful fashion
7	Providing an efficient search feature
8	Getting good feedback from Michael Chen and other senior consultants
9	Effectively promoting the new system
10	Realizing the benefits of the new system within one year

Case Study: Predictive Approach

Project Execution

- Usually takes the most resources to perform
  - Project managers must use their leadership skills to handle the many challenges that occur during project execution
- Lists the knowledge areas, executing processes, and outputs of project execution
  - Many project sponsors and customers focus on deliverables related to providing the products, services, or results desired from the project
  - It is equally important to document change requests and update planning documents
- A milestone report can help focus on completing major milestones

Case Study: Predictive Approach

Project Monitoring and Controlling

- Involves measuring progress toward project objectives, monitoring deviation from the plan, and taking corrective actions
  - Affects all other process groups and occurs during all phases of the project life cycle
- Outputs include performance reports, change requests, and updates to various plans

Case Study: Predictive Approach

Project Closing

- Involves gaining stakeholder and customer acceptance of the final products and services
  - Even if projects are not completed, they should be closed out to learn from the past
- Outputs may include project files and lessons-learned reports
  - Also may include a final report and presentation

Case Study: Agile Approach

JWD Consulting's Project Management Intranet Site

- An agile project team typically uses several iterations or deliveries of software instead of waiting until the end of the project to provide one product

Case Study: Agile Approach

***Scrum Roles**, Artifacts, and Ceremonies*

- **Product owner**: person responsible for the business value of the project and for deciding what work to do and in what order -> product backlog
- **ScrumMaster**: person who ensures that the team is productive, facilitates the daily Scrum, enables close cooperation across all roles and functions, and removes barriers that prevent the team from being effective
- **Scrum team or development team**: cross-functional team of five to nine people who organize themselves and the work to produce the desired results for each sprint, which normally lasts two to four weeks

Case Study: Agile Approach

*Scrum Roles, **Artifacts**, and Ceremonies*

- An **artifact** is a useful object created by people
- Scrum artifacts
  - **Product backlog**: list of features prioritized by business value
  - **Sprint backlog**: highest-priority items from the product backlog to do in a sprint
  - **Burndown chart**: shows the cumulative work remaining in a sprint on a day-by-day basis

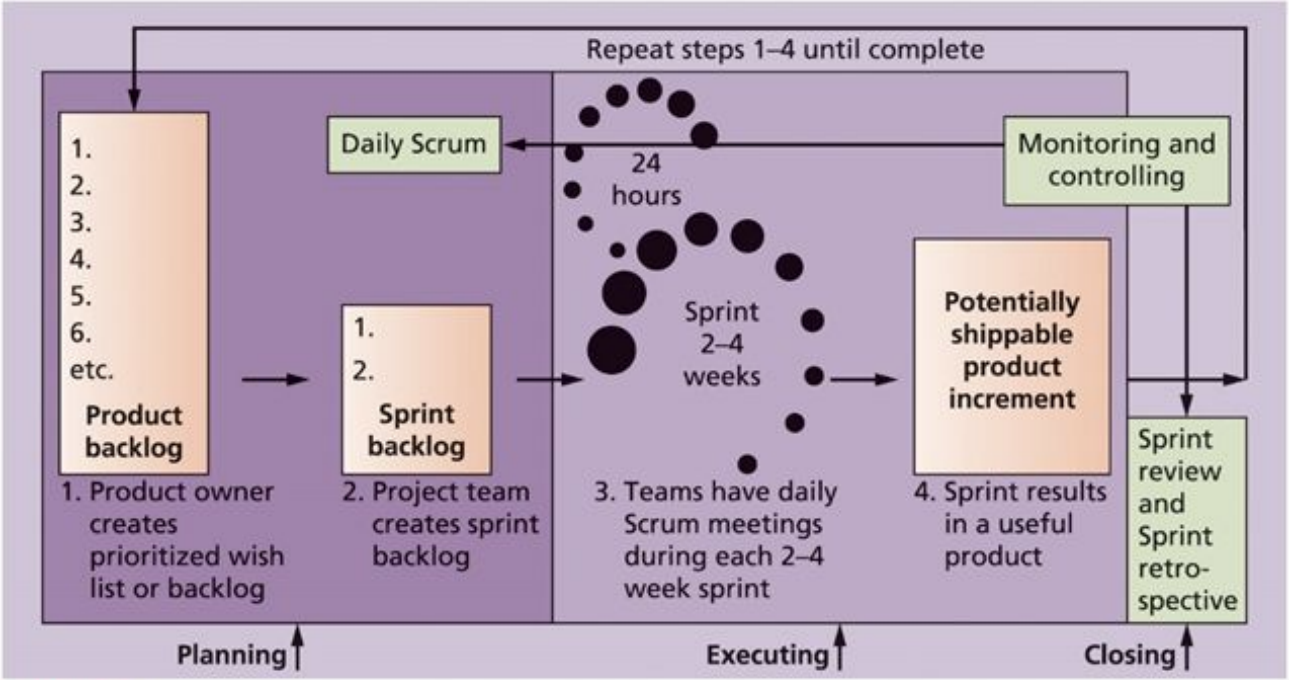
Case Study: Agile Approach

*Scrum Roles, Artifacts, and **Ceremonies***

- Scrum ceremonies
  - **Sprint planning session**: meeting with the team to select a set of work from the product backlog to deliver during a sprint
  - **Daily Scrum**: short meeting for the development team to share progress and challenges and plan work for the day
  - **Sprint reviews**: meeting in which the team demonstrates to the product owner what it has completed during the sprint
  - **Sprint retrospectives**: meeting in which the team looks for ways to improve the product and the process based on a review of the actual performance of the development team

Case Study: Agile Approach

Scrum Roles, Artifacts, and Ceremonies



Scrum framework and the process groups

Case Study: Agile Approach

Scrum Roles, Artifacts, and Ceremonies

Process Group	Scrum Activity
Initiating	Determine roles
	Decide how many sprints will compose each release and the scope of software to deliver
Planning	Create product backlog
	Create sprint backlog
	Create release backlog
	Plan work each day in the daily Scrum
	Document stumbling blocks in a list
Executing	Complete tasks each day during sprints
	Produce a shippable product at the end of each sprint

Case Study: Agile Approach

Pre-Initiation and Initiation

- Main differences between pre-initiation in this case and the first case
  - Determining roles and deciding what functionality would be delivered for each release
  - How many sprints will be required to complete a release
  - How many releases of software to deliver

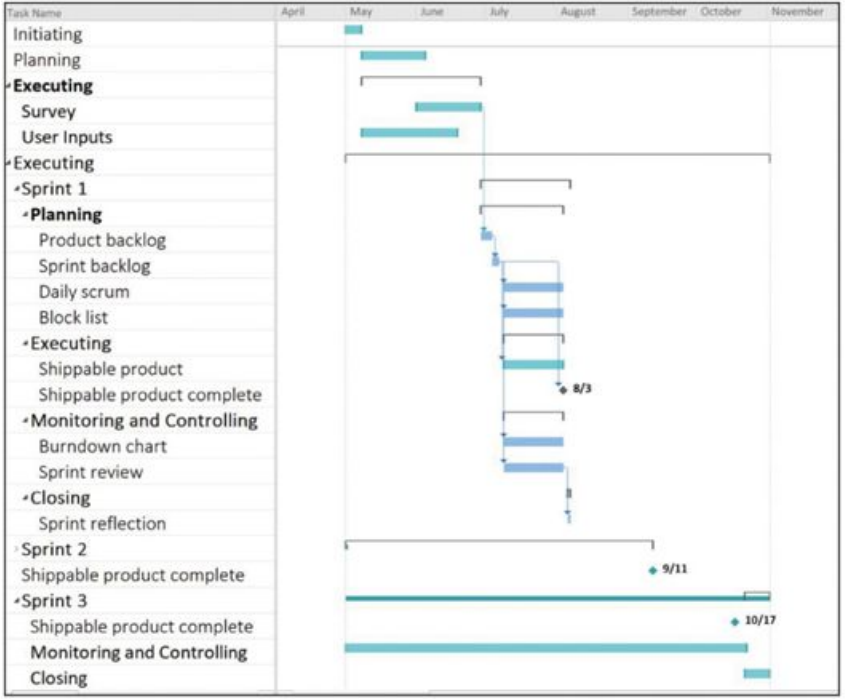
Case Study: Agile Approach

Planning

- Because Scrum implies that team members work as a self-directed group, coached by the ScrumMaster, a **team charter** should not be necessary
- Descriptions of work are identified in the product and sprint backlogs
- More detailed work is documented in technical stories
- Team must estimate a velocity or capacity for each sprint

Case Study: Agile Approach

Planning



Intranet site project baseline Gantt chart using Scrum approach

Case Study: Agile Approach

Planning

Product Backlog	Sprint Backlog
1. User story templates, samples, and point person	1. User story templates, samples, and point person
2. WBS templates, samples, and point person	2. WBS templates, samples, and point person
3. Project schedule templates, samples, and point person	3. Project schedule templates, samples, and point person
4. Ability to charge customers for some intranet products and services	4. Ability to charge customers for some intranet products and services
5. Ability to collect user suggestions	5. Ability to collect user suggestions
6. Business case templates, samples, and point person	
7. Ask the Expert feature	
8. Stakeholder management strategy templates, samples, and point person	
9. Risk register templates, samples, and point person	
10. Etc.	

Case Study: Agile Approach

Executing

- The most time and money should be spent on executing
  - Plans are implemented to create the desired product
- Agile approach: team produces several iterations of a potentially shippable product
  - Users can access and make suggestions
- Communications are different
  - Project team meets every morning, physically or virtually

Case Study: Agile Approach

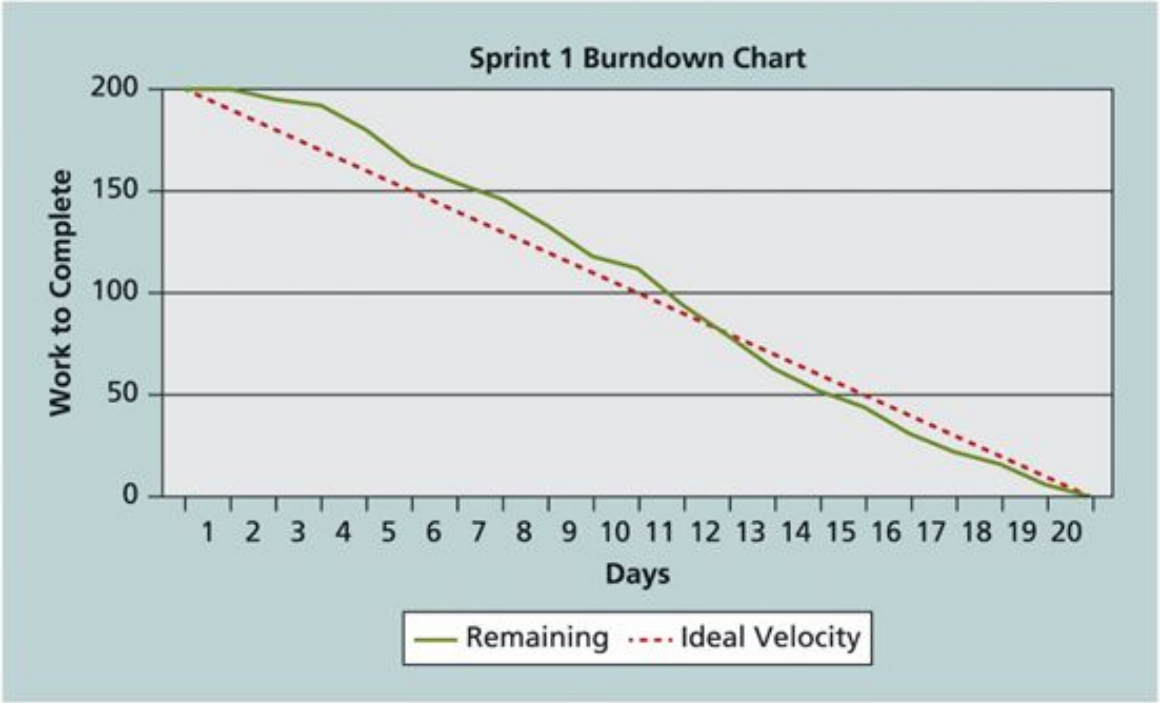
Monitoring and Controlling

- The two main tools for monitoring and controlling in the Scrum framework
  - **Daily Scrum**: held each morning to plan and communicate work for the day and discuss any risks, issues, or blockers
  - **Sprint review**: work progress within a sprint can be represented on a sprint board maintained by the ScrumMaster

    Burndown chart: an important artifact used to graphically display progress on each sprint

Case Study: Agile Approach

Monitoring and Controlling



Case Study: Agile Approach

Close

- After the sprint **review**, the ScrumMaster leads a sprint **retrospective**
  - Team reflects on what happened during the sprint
- Sprint retrospective is intended to answer two fundamental questions
  - What went well during the last sprint that we should continue doing?
  - What could we do differently to improve the product or process?